

Exploration Drones

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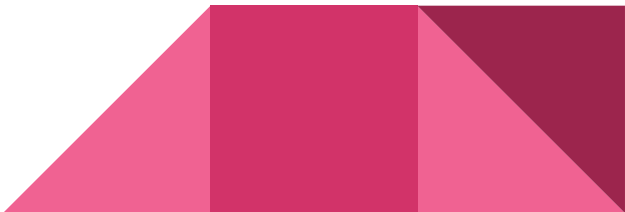
Introduction

A group of drones want to explore and map a grid completely

Drones communicate to ensure they don't double-cover area

To ensure the communication is private between the drones, each drone would have an encrypted key which would be verified if it is a valid drone.

Because of this pairwise communication will be used where one drone can only talk to one other drone at a time.

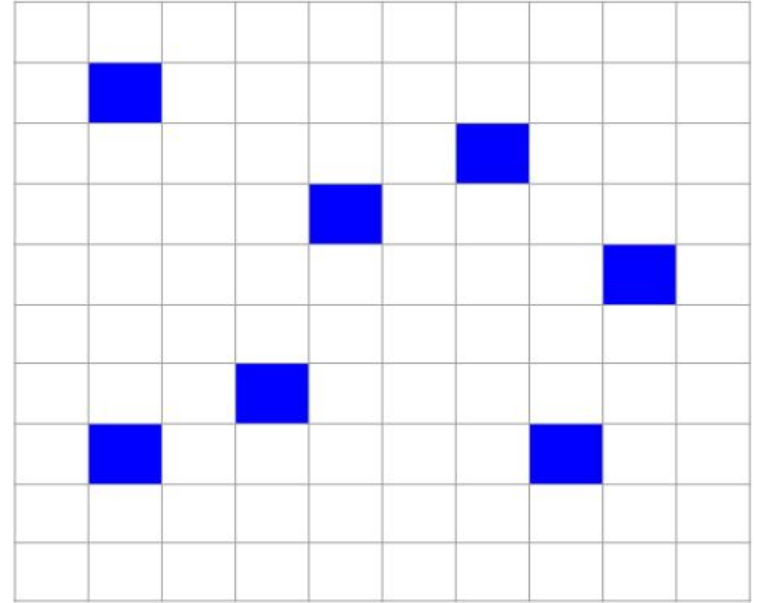


Map

Map is a 10 by 10 grid and the goal is to explore the entire map using the least amount of energy on communication.

White squares(unexplored area)

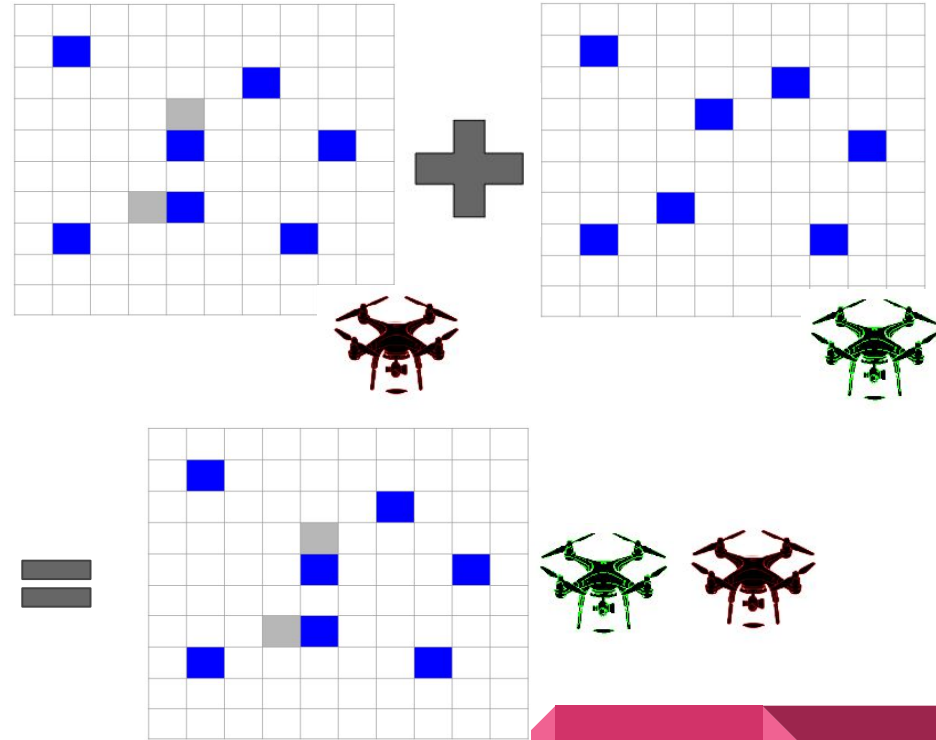
Blue squares (drone positions)



Drone communication

Each Drone has a local version of the map and a transmitting drone(red) can transmit their map to another drone(green) and the receiver drone can merge it

Drones need to communicate frequently so that all the drones have updated maps





Balanced Incomplete Block Designs

BIBD is a way to arrange many items (points) into smaller groups (blocks) while ensuring fairness: every item appears equally often (r times), and every pair of items appears together in the same block the same number of times (λ times)

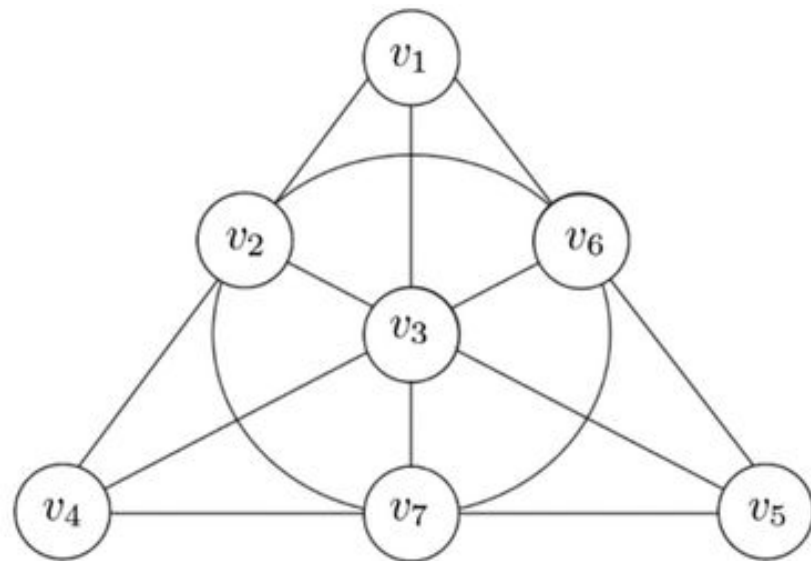
(v, b, r, k, λ)

- V number of points
- B number of blocks
- Each point lies in r blocks
- Each block contains k points
- Every pair of distinct points appears together in λ blocks* $\rightarrow$ BALANCED

$$r = \frac{\lambda(v-1)}{k-1}$$

$$b = \frac{vr}{k} = \frac{\lambda(v^2-v)}{k^2-k}$$

$$\begin{array}{c}
 \ell_1 \\
 \ell_2 \\
 \ell_3 \\
 \ell_4 \\
 \ell_5 \\
 \ell_6 \\
 \ell_7
 \end{array}
 \begin{bmatrix}
 v_1 & v_2 & v_3 & v_4 & v_5 & v_6 & v_7 \\
 1 & 1 & 0 & 1 & 0 & 0 & 0 \\
 0 & 1 & 1 & 0 & 1 & 0 & 0 \\
 0 & 0 & 1 & 1 & 0 & 1 & 0 \\
 0 & 0 & 0 & 1 & 1 & 0 & 1 \\
 1 & 0 & 0 & 0 & 1 & 1 & 0 \\
 0 & 1 & 0 & 0 & 0 & 1 & 1 \\
 1 & 0 & 1 & 0 & 0 & 0 & 1
 \end{bmatrix}$$



**BIBDs are highly regular incidence
structures!**

Difference Sets

Find the other blocks of the $(7,3,1)$ -BIBD using the “starter” block $D = \{1,2,4\}$.

1. Is this possible?
2. What is the relationship between D and BIBDs?
3. How are blocks within a BIBD found from known ones?

$$D = \{1, 2, 4\}, \quad G = \mathbb{Z}_7 = \{0, 1, 2, 3, 4, 5, 6\}$$

$$d_1 - d_2 \pmod{7}$$

$$D = \{1, 2, 4\}, \quad G = \mathbb{Z}_7 = \{0, 1, 2, 3, 4, 5, 6\}$$

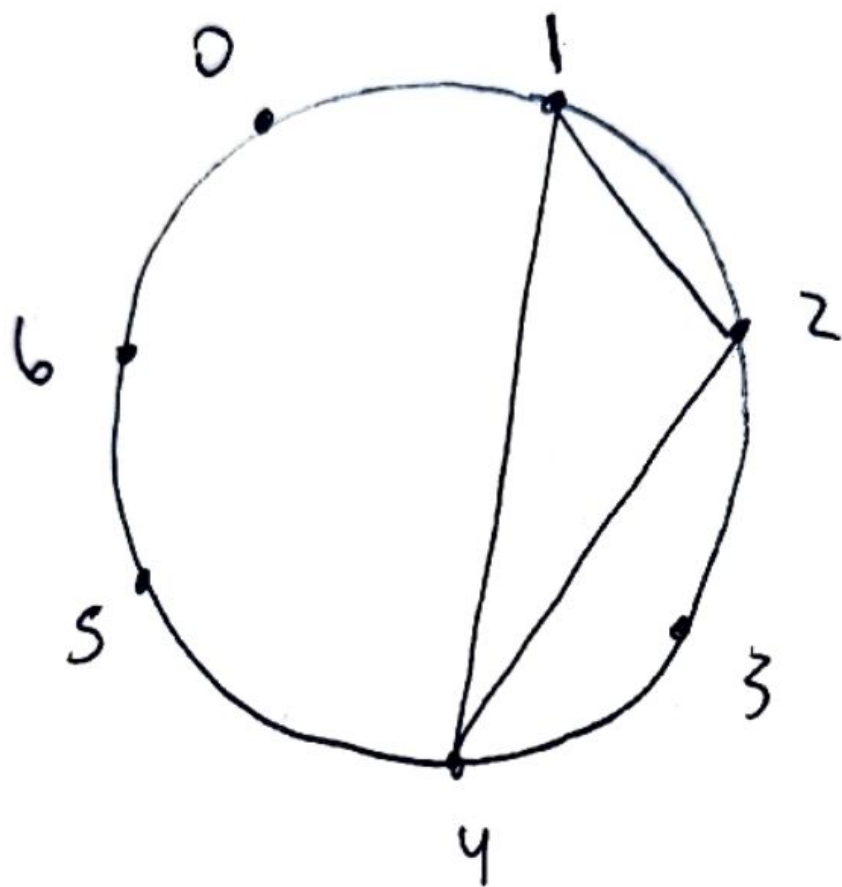
d_1	d_2	$d_1 - d_2 \pmod{7}$
1	2	6
1	4	4
2	1	1
2	4	5
4	1	3
4	2	2

$$D+g = \{d+g \pmod{7} : d \in D\}$$

J	Block $D+g$
0	$\{4, 2, 4\}$
1	$\{2, 3, 5\}$
2	$\{3, 4, 6\}$
3	$\{4, 5, 0\}$
4	$\{5, 6, 1\}$
5	$\{6, 0, 2\}$
6	$\{0, 1, 3\}$

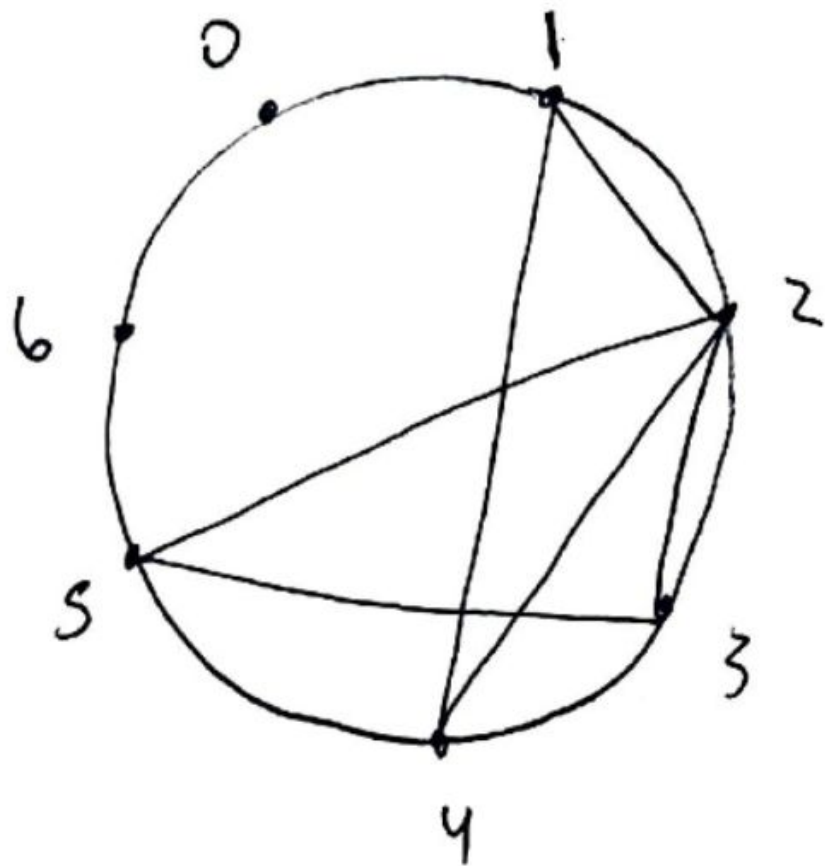
$$D+g = \{d+g \pmod{7} : d \in D\}$$

D	Block $D+g$
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1	$\{2, 3, 5\}$
2	$\{3, 4, 6\}$
3	$\{4, 5, 0\}$
4	$\{5, 6, 1\}$
5	$\{6, 0, 2\}$
6	$\{0, 1, 3\}$



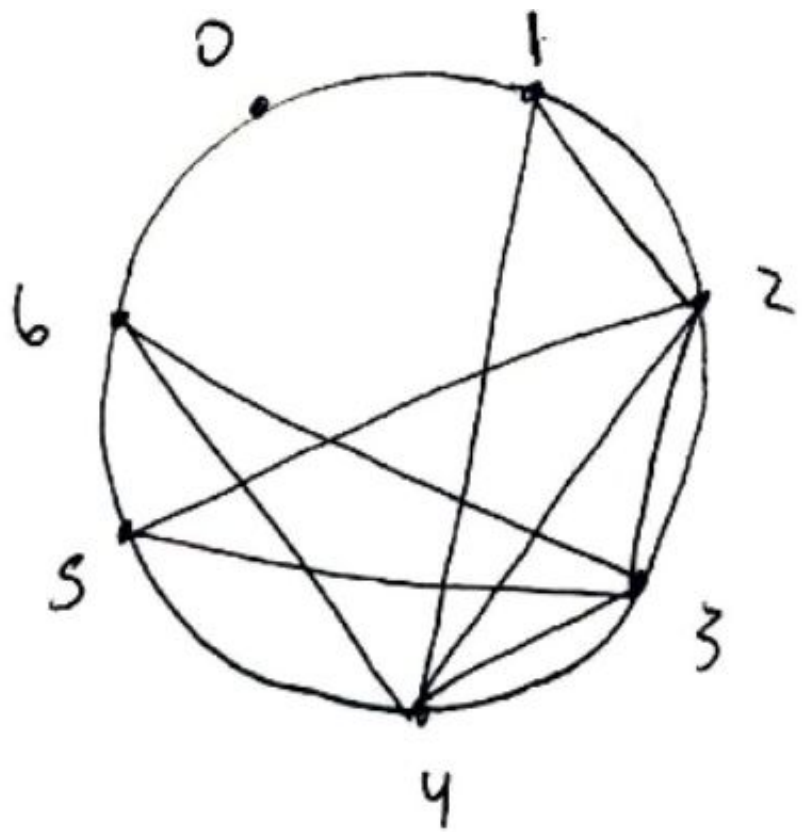
$$D+g = \{d+g \pmod{7} : d \in D\}$$

D	Block $D+g$
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1	$\{2, 3, 5\}$
2	$\{3, 4, 6\}$
3	$\{4, 5, 0\}$
4	$\{5, 6, 1\}$
5	$\{6, 0, 2\}$
6	$\{0, 1, 3\}$



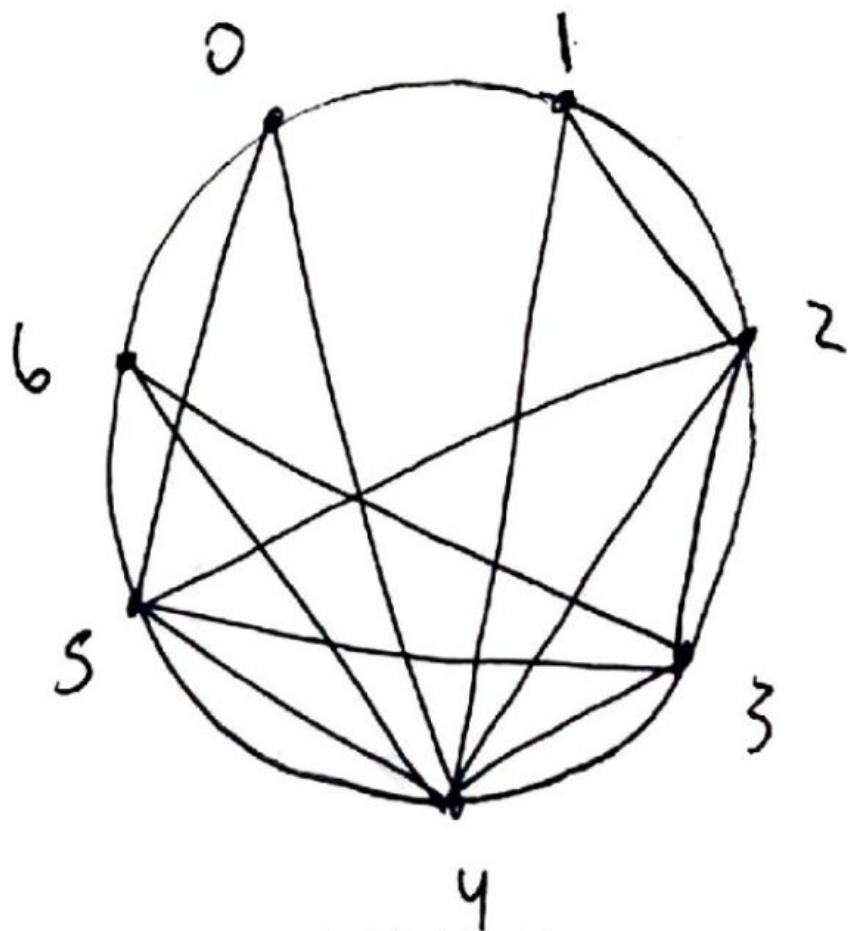
$$D+g = \{d+g \pmod{7} : d \in D\}$$

g	Block $D+g$
0	$\{4, 2, 4\}$
1	$\{2, 3, 5\}$
2	$\{3, 4, 6\}$
3	$\{4, 5, 0\}$
4	$\{5, 6, 1\}$
5	$\{6, 0, 2\}$
6	$\{0, 1, 3\}$



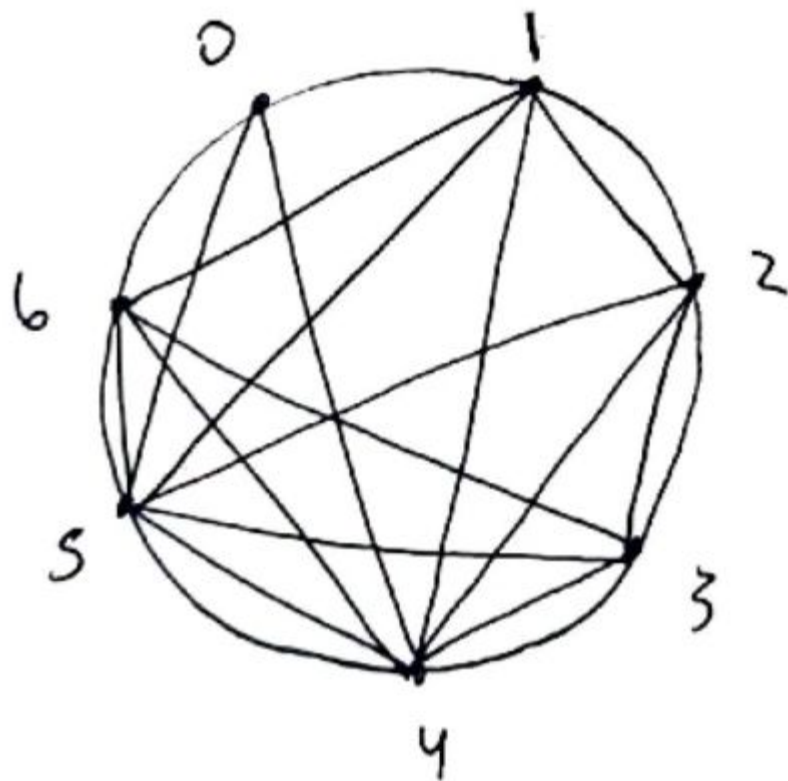
$$D+g = \{d+g \pmod{7} : d \in D\}$$

D	Block $D+g$
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1	$\{2, 3, 5\}$
2	$\{3, 4, 6\}$
3	$\{4, 5, 0\}$
4	$\{5, 6, 1\}$
5	$\{6, 0, 2\}$
6	$\{0, 1, 3\}$



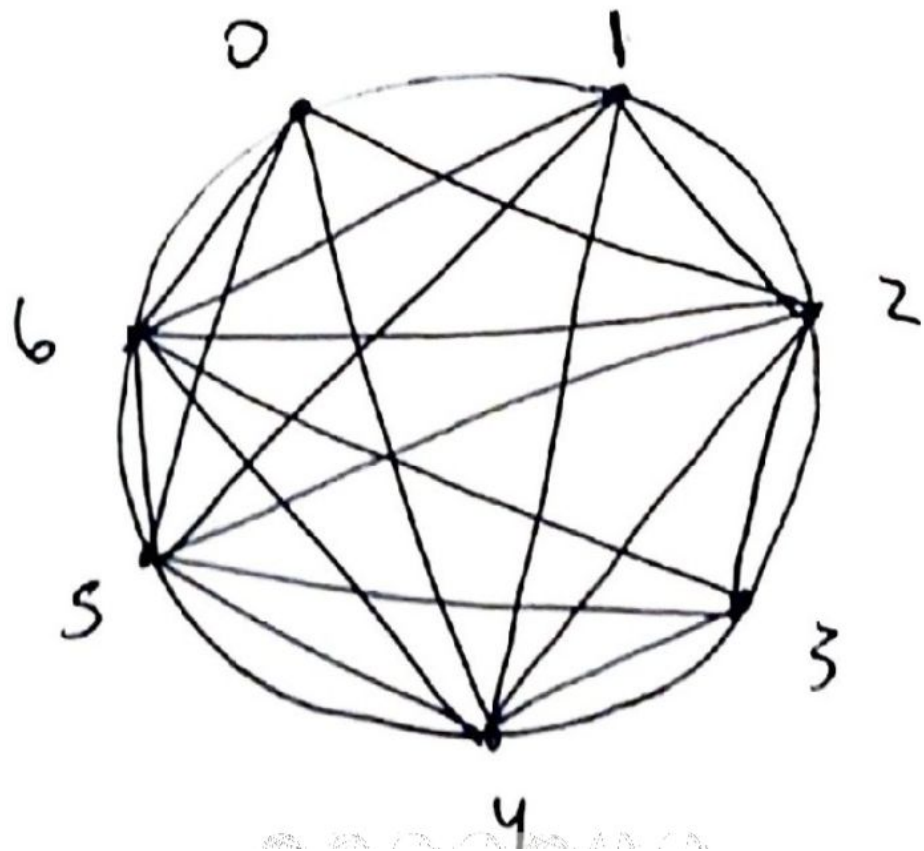
$$D+g = \{d+g \pmod{7} : d \in D\}$$

g	Block $D+g$
0	$\{4, 2, 4\}$
1	$\{2, 3, 5\}$
2	$\{3, 4, 6\}$
3	$\{4, 5, 0\}$
4	$\{5, 6, 1\}$
5	$\{6, 0, 2\}$
6	$\{0, 1, 3\}$



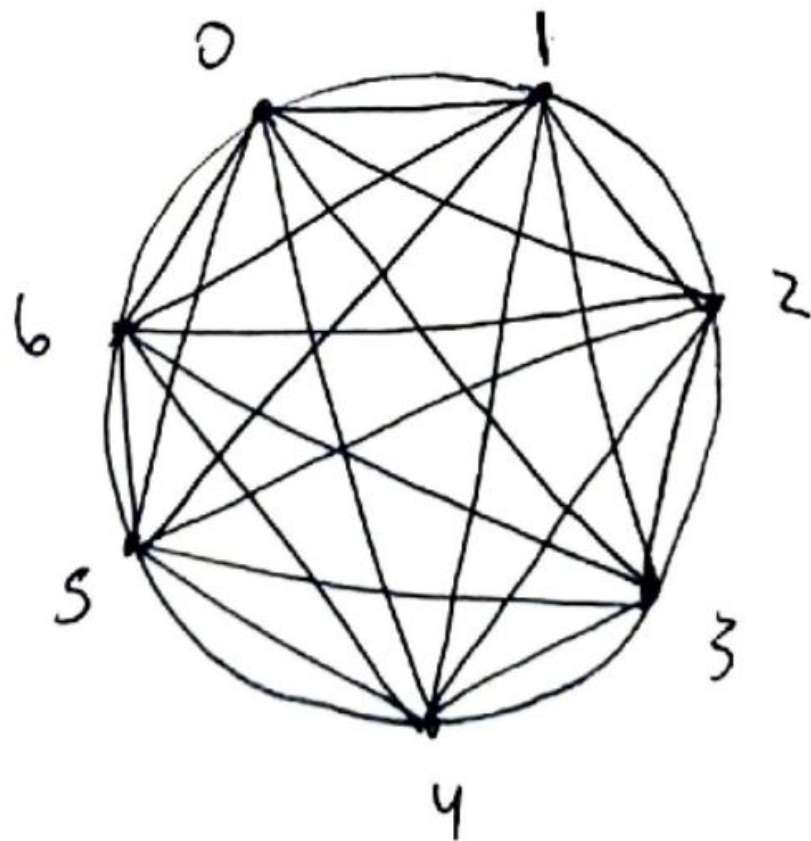
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$$D+g = \{d+g \pmod{7} : d \in D\}$$

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3	$\{4, 5, 0\}$
4	$\{5, 6, 1\}$
5	$\{6, 0, 2\}$
6	$\{0, 1, 3\}$

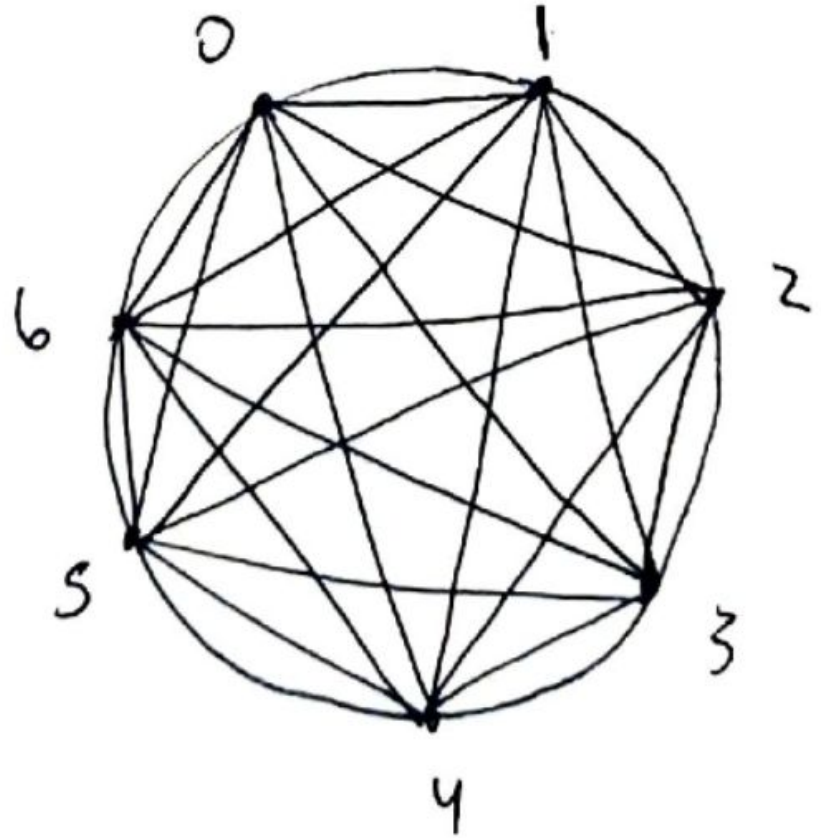


(7,3,1) difference set.

- 7 points
- 7 blocks
- 3 points/block
- 3 blocks/point
- Distinct pair in 1 block
- Full parameters: $(7,7,3,3,1)$

Every translation corresponds with every difference $\rightarrow$ balance condition

Difference sets make symmetric BIBDs*



**Difference sets are algebraic structures
that can be used to find symmetric BIBDs!**

Projective Geometry

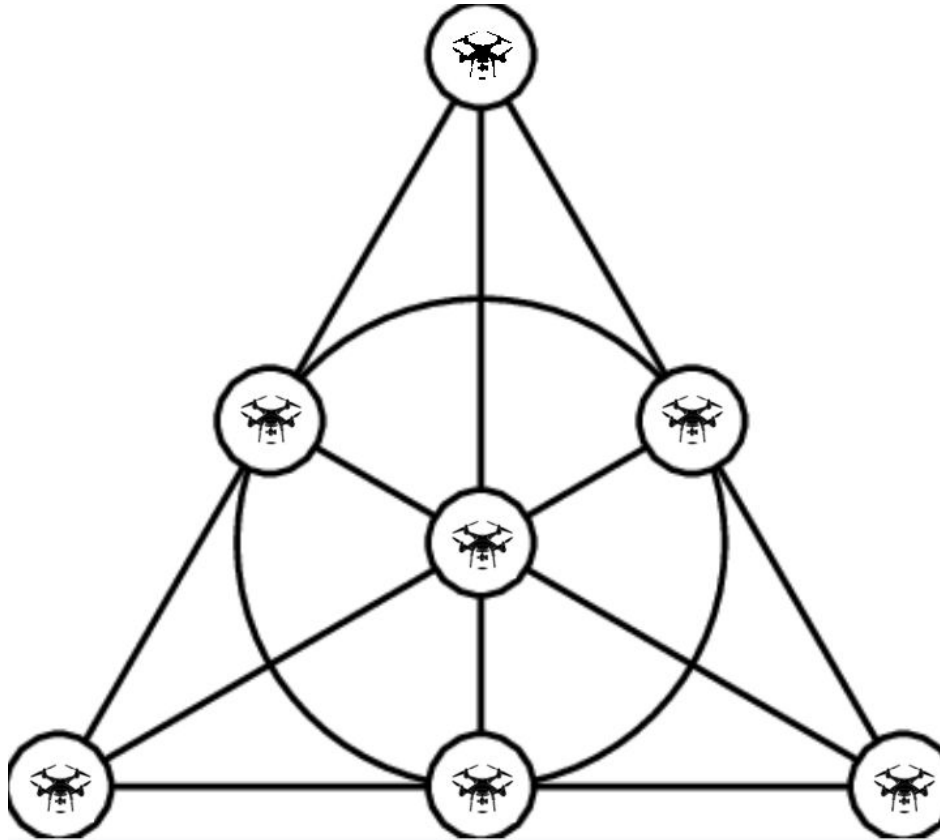
**A projective
plane (P, L, I) is
an incidence
structure.**

1. Any two distinct points lie on exactly one line.
2. Any two distinct lines intersect at exactly one point.
3. There are four points such that no three are collinear.
(nondegeneracy)

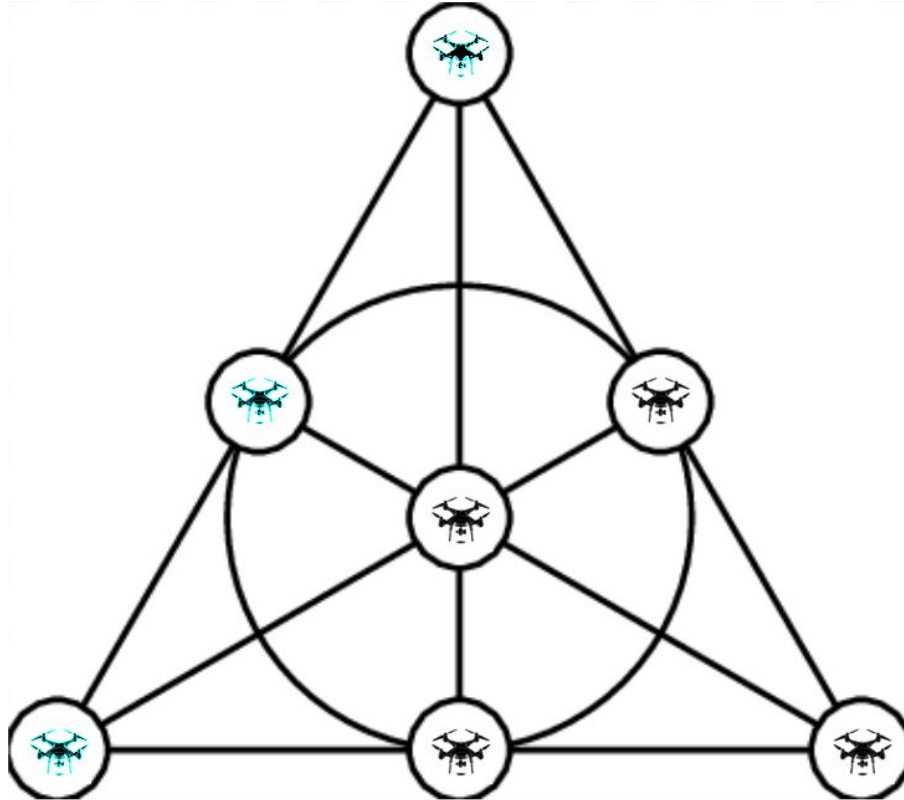


BIBD to Facilitate Drone Communication

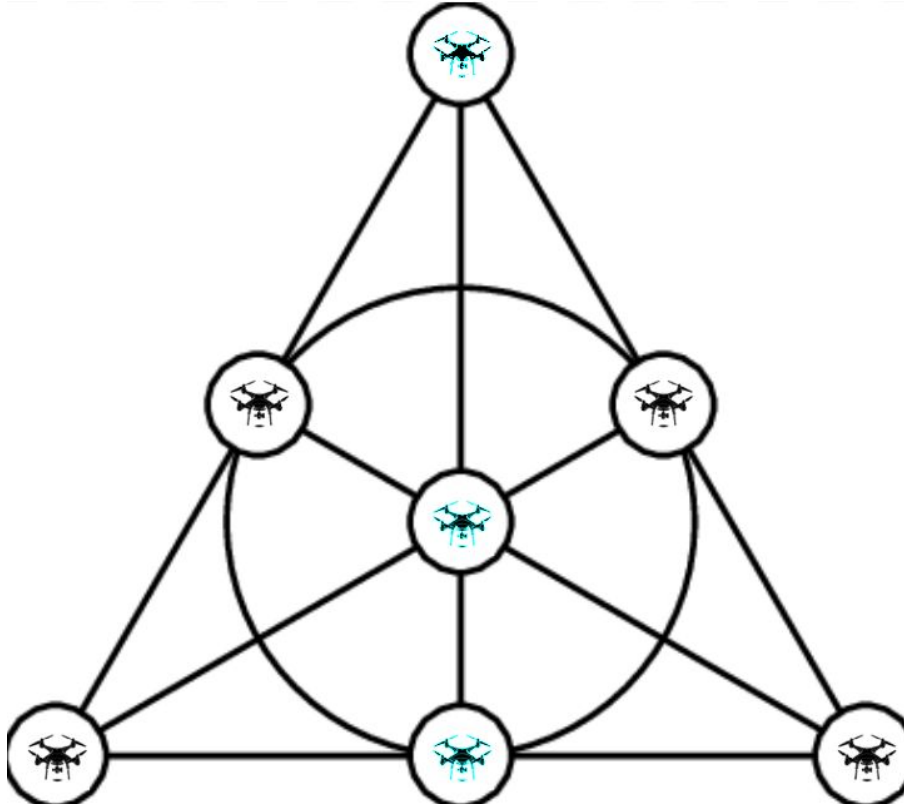
7 3 1 Applied to Drones



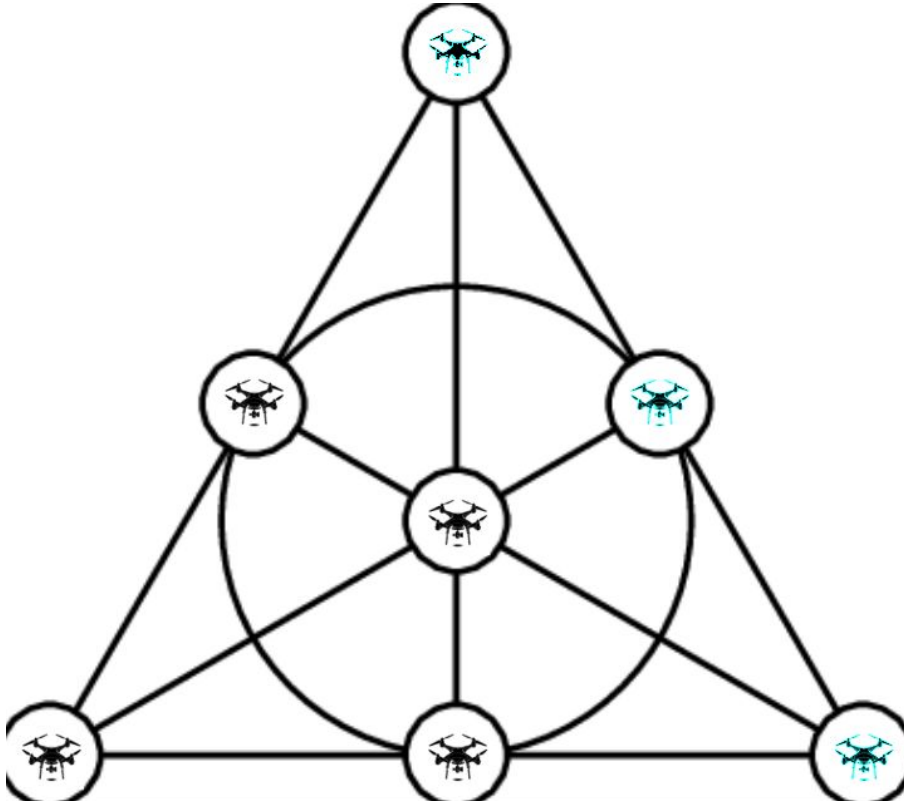
7 3 1 Applied to Drones



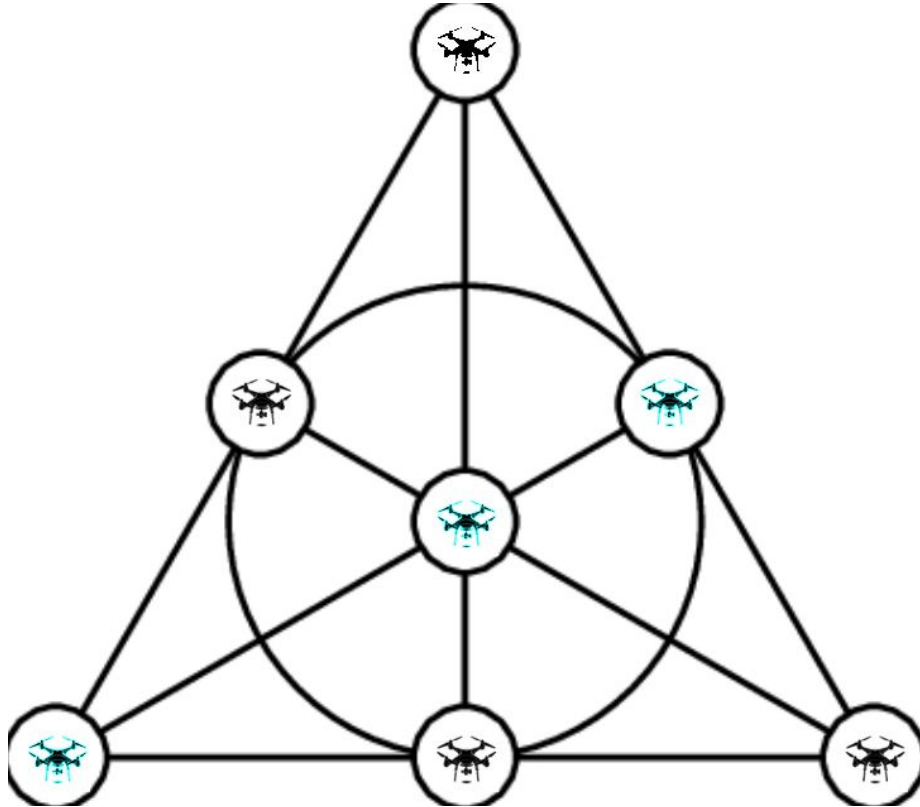
7 3 1 Applied to Drones



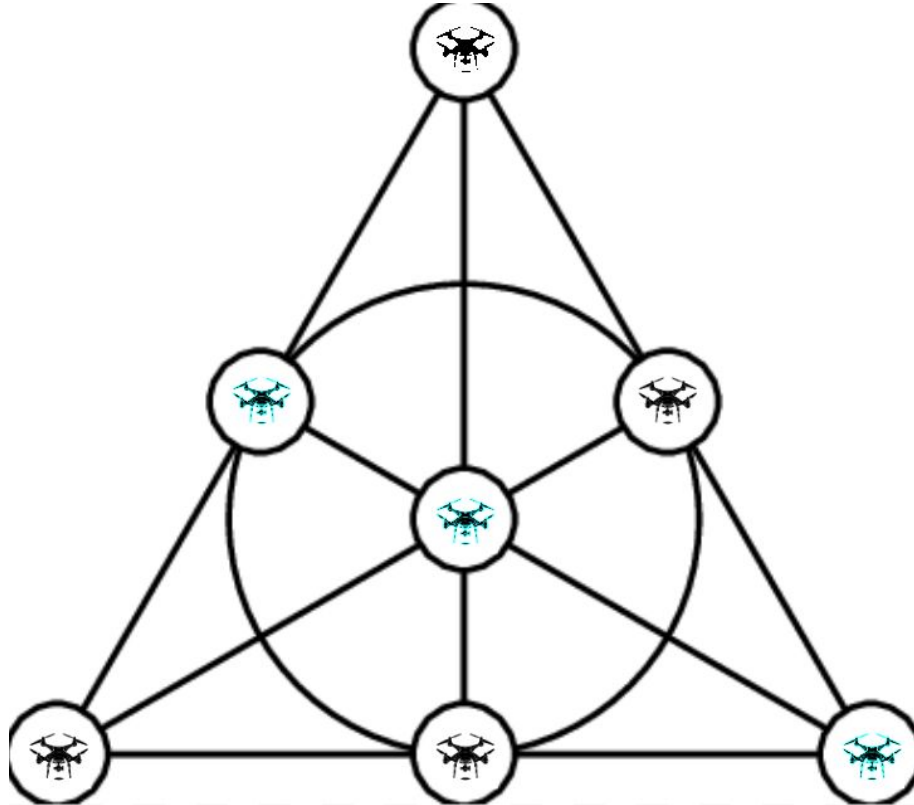
7 3 1 Applied to Drones



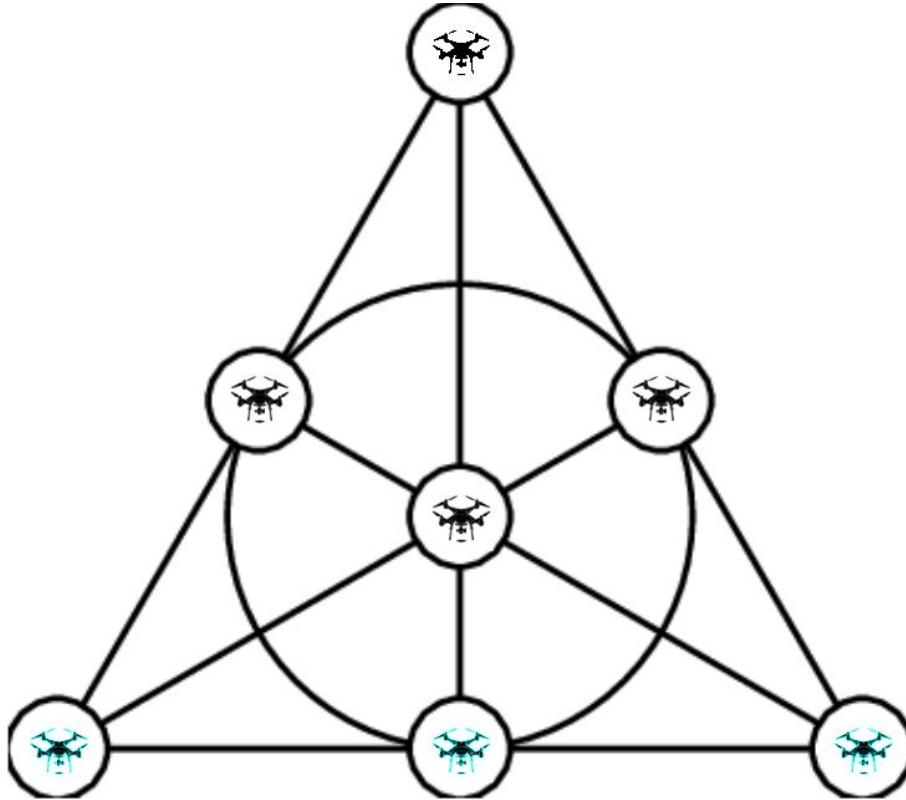
7 3 1 Applied to Drones



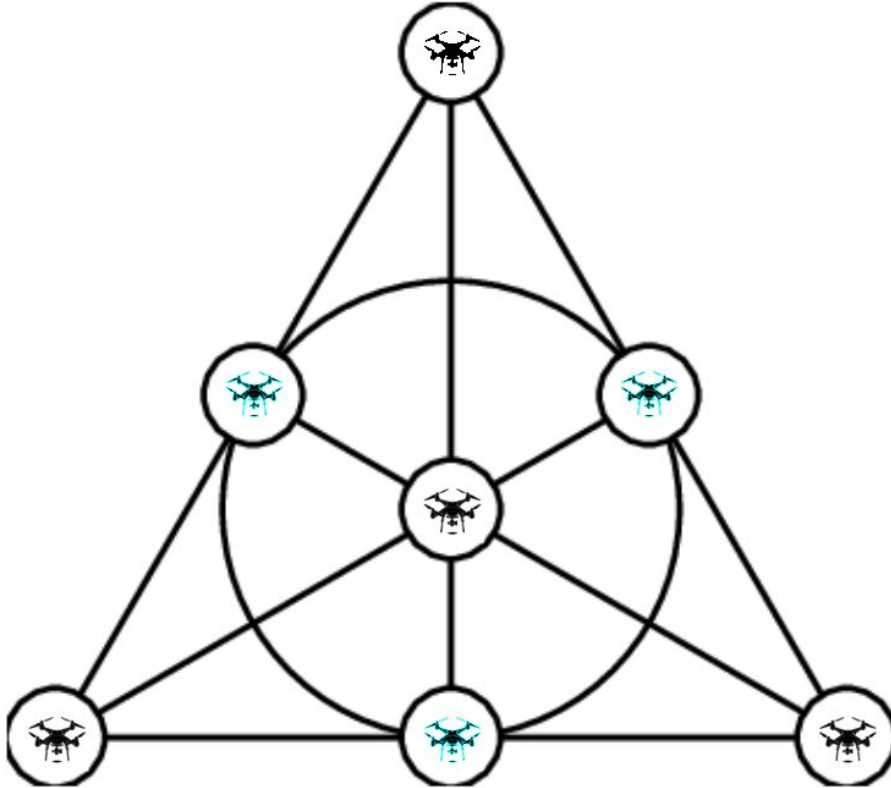
7 3 1 Applied to Drones



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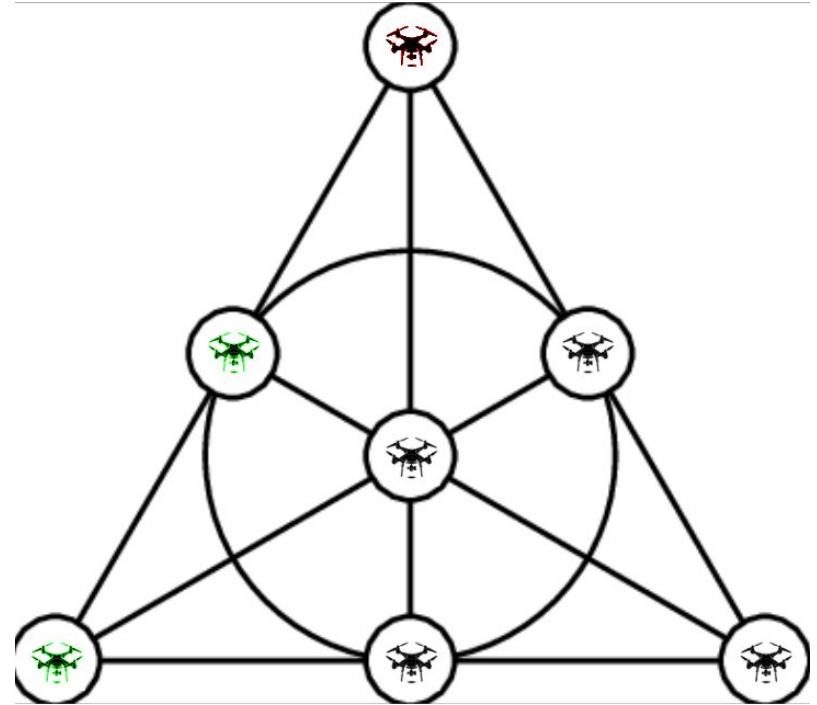
7 3 1 Applied to Drones



7 3 1 Applied to Drones

Communication within a group happens first. Since every group is connected to every other group, this ensures every drone is included in communication.

Receivers and transmitters on drones not in a group are not on while other drones are communicating



731 difference set.

The 7 3 1 difference set can be used to create schedules for the drones.

groups differ by 1 (mod 7). So if you add 1 (mod 7) 7 times you can get 7 different groups that are difference sets.

7 sets- each number (0-6) appears exactly 3 times.

Each group is connected to every other group

[0, 1, 3]

[1, 2, 4]

[2, 3, 5]

[3, 4, 6]

[0, 4, 5]

[1, 5, 6]

[0, 2, 6]

Communication schedules

Blue squares are the drones groups that are communicating at a time.

White squares are drones that are not communicating during that time.

Communication is pairwise so every permutation between the 3 drones would be done before moving on

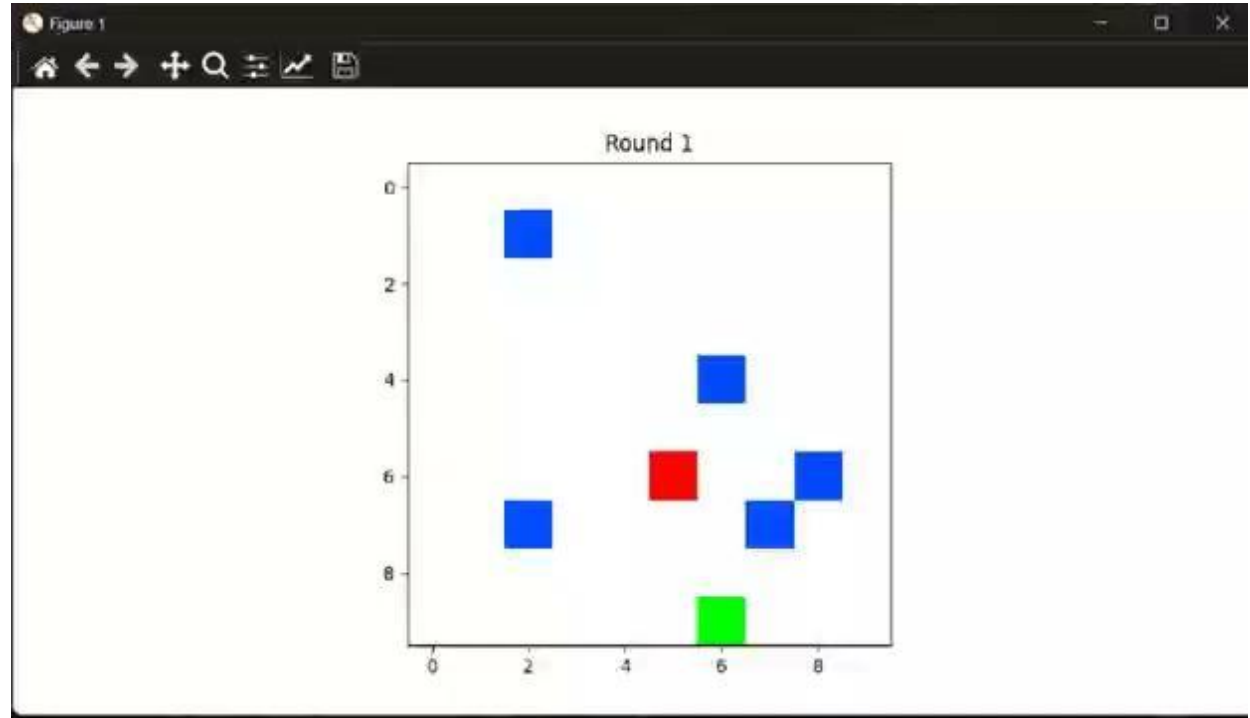
0	Blue	White	White	White	Blue	White	Blue
1	Blue	Blue	White	White	White	Blue	White
2	White	Blue	Blue	White	White	White	Blue
3	Blue	White	Blue	Blue	White	White	White
4	White	Blue	White	Blue	Blue	White	White
5	White	White	Blue	White	Blue	Blue	White
6	White	White	White	Blue	White	Blue	Blue

Simulated Exploration

Communication without BIBD

Every drone transmits to every drone.

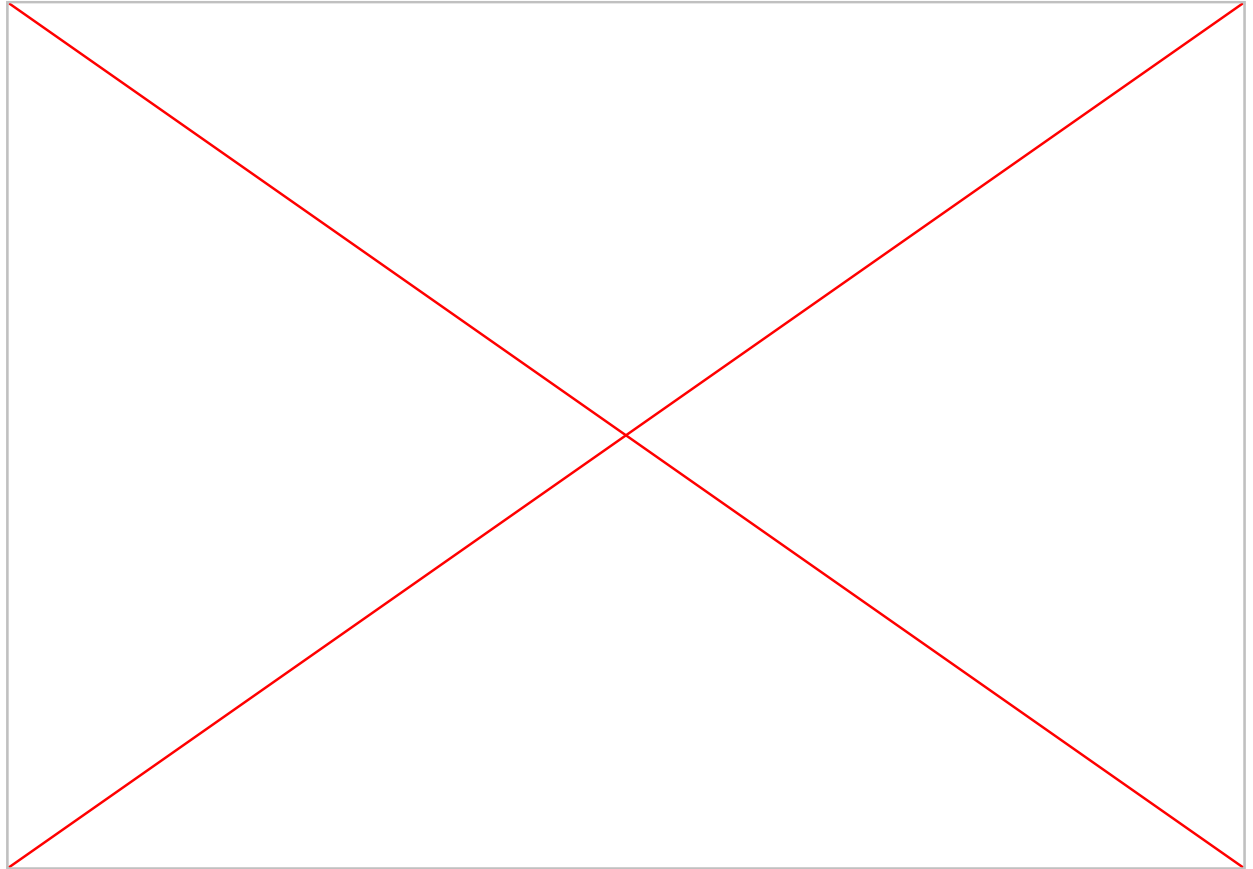
1806 communications



Communication with BIBD

Communication is
scheduled by difference
sets

306 communications



Energy Consumption of BIBD/Non BIBD

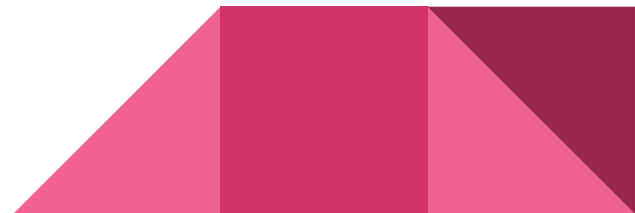
Energy consumption

Drone State	Send	Receive	Idle	Sleep
Energy (W)	1.4	1	0.83	0.13

(Source: <https://www.sigmobility.org/mobihoc/2003/papers/p35-zheng.pdf>)

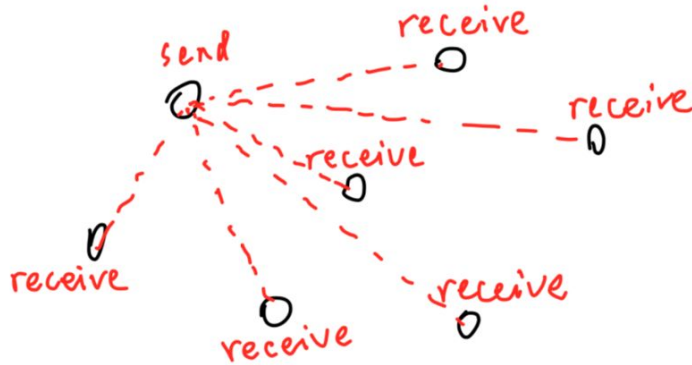
Two ways of communication:

- Public: every drone broadcasts public data that can be received by all active drones
- Pairwise: drones can only communicate one-to-one



Energy for public communication

- Non-BIBD, (7, 3, 1) example:



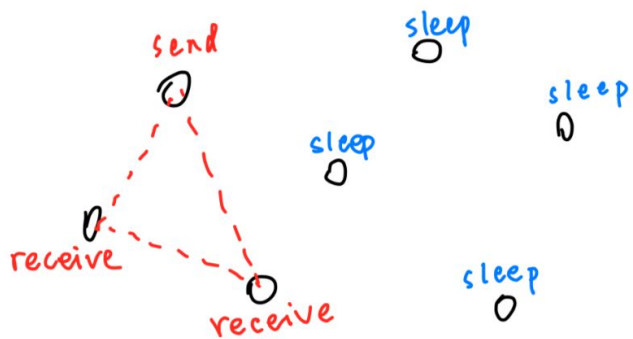
7 rounds comm.

$$(\text{send} + 6 \text{ receive}) \times 7$$

$$= 7 \text{ send} + 42 \text{ receive}$$

Energy for public communication

- BIBD, (7, 3, 1) example:



7 blocks $\rightarrow$ 7 rounds comm.

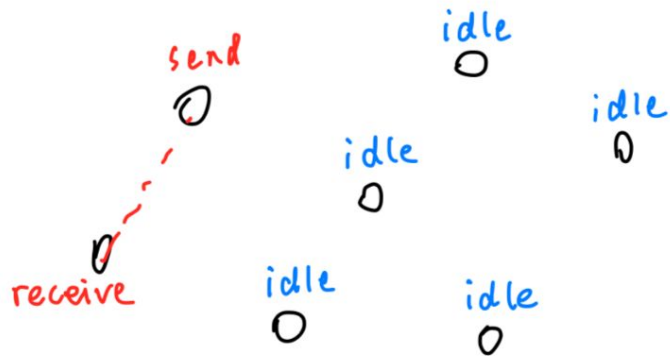
3 rounds comm. per block

$$(\text{send} + 2 \text{ receive} + 4 \text{ sleep}) \times 3 \times 7$$

$$= 21 \text{ send} + 42 \text{ receive} + 84 \text{ sleep}$$

Energy for pairwise communication

- Non-BIBD, (7, 3, 1) example:



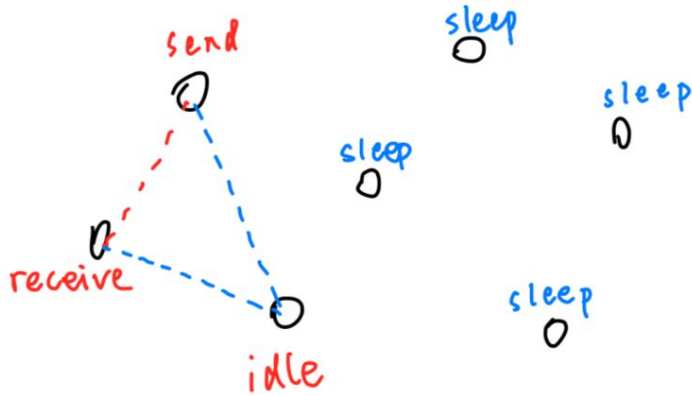
$$\binom{7}{2} = 21 \text{ rounds comm.}$$

$$(\text{send} + \text{receive} + 5 \text{ idle}) \times 21 \times 2$$

$$= 42 \text{ send} + 42 \text{ receive} + 210 \text{ idle}$$

Energy for pairwise communication

- BIBD, (7, 3, 1) example:



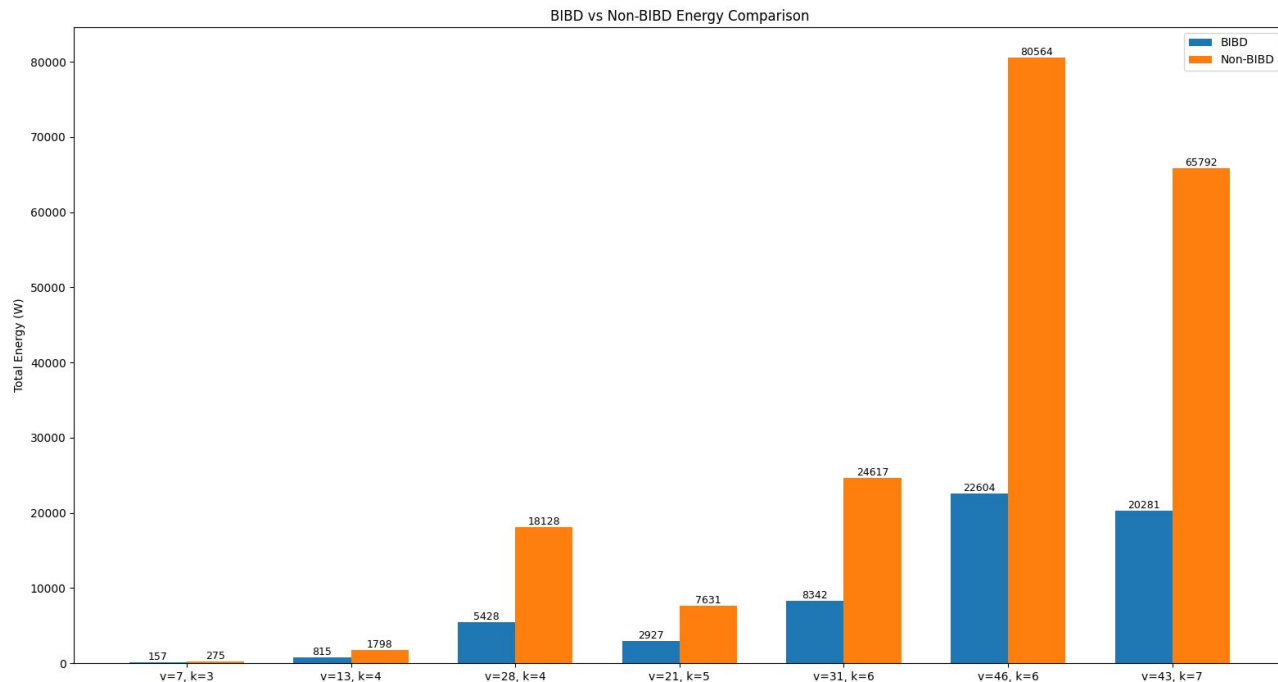
7 blocks $\Rightarrow$ 7 rounds comm.

b rounds comm. per block

$$(send + receive + idle + 4 sleep) \times b \times 7$$

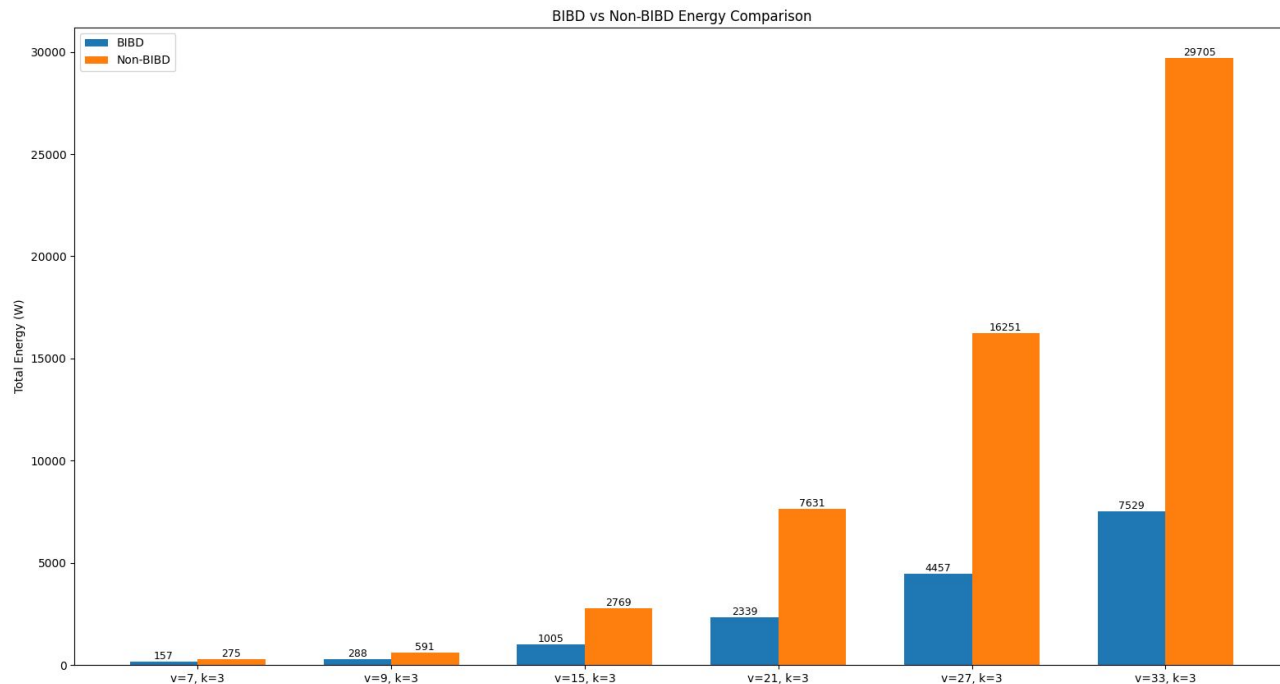
$$= 42send + 42receive + 42idle + 168sleep$$

Energy consumption – classic BIBD sets



(7, 3, 1),
(13, 4, 1),
(28, 4, 1),
(21, 5, 1),
(31, 6, 1),
(46, 6, 1),
(43, 7, 1)

Energy consumption – same k



(7, 3, 1),
(9, 3, 1),
(15, 3, 1),
(21, 3, 1),
(27, 3, 1),
(33, 3, 1)

Conclusion

Conclusion

BIBDs were more energy efficient than the non BIBD approach and this approach scaled well when more drones were added when drones did pairwise communication

If a valid BIBD can be made from the number of drones being used, a BIBD strategy for communication schedules would be more energy efficient.



Future Work

Future Works

- Test other types of BIBDs on this problems like unsymmetric BIBDs
- Add a bad actor drone and test if a combinatorial approach performs well when a bad actor must be determined and excluded from communication.

